many thanks to Kevin Wayne for this nifty assignment

This assignment is from http://nifty.stanford.edu/

This programming assignment will give you practice with queues, interfaces, objects, and arrays of objects. You are going to implement two classes that allow us to simulate a guitar. We will be using two utility classes known as StdAudio and StdDraw that are used in the Princeton intro CS course. You don't have to understand the details of these utility classes, but if you are interested, you can read about them at the following url:

http://introcs.cs.princeton.edu/java/stdlib/

When a guitar string is plucked, the string vibrates and creates sound. The length of the string determines its fundamental frequency of vibration. We model a guitar string by sampling its displacement (a real number between -1/2 and +1/2) at N equally spaced points (in time), where N equals the sampling rate (44,100) divided by the fundamental frequency (rounded to the nearest integer). We store these displacement values in a structure that we will refer to as a ring buffer.



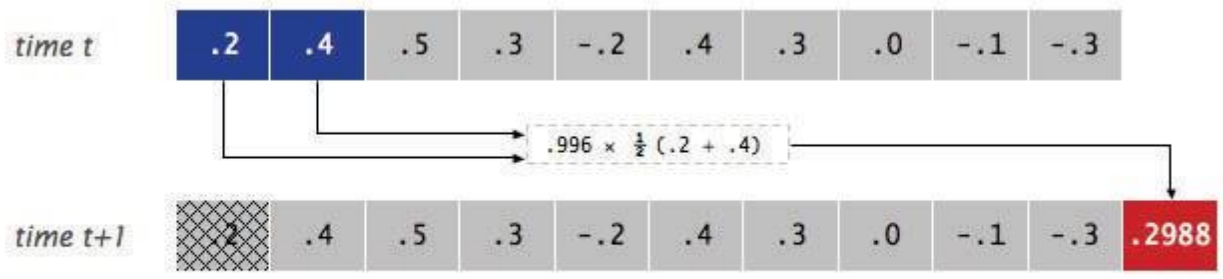
Plucking the string

The excitation of the string can contain energy at any frequency. We simulate the excitation by filling the ring buffer with white noise: set each of the N sample displacements to a random real number between -1/2 and +1/2.



The resulting vibrations

After the string is plucked, the string vibrates. The pluck causes a displacement which spreads wave-like over time. The Karplus-Strong algorithm simulates this vibration by maintaining a ring buffer of the N samples: for each step the algorithm deletes the first sample from the ring buffer and adds to the end of the ring buffer the average of the first two samples, scaled by an energy decay factor of 0.996.



the Karplus-Strong update

Why it works

The two primary components that make the Karplus-Strong algorithm work are the ring buffer feedback mechanism and the averaging operation.

- The ring buffer feedback mechanism: The ring buffer models the medium (a string tied down at both ends) in which the energy travels back and forth. The length of the ring buffer determines the fundamental frequency of the resulting sound. Sonically, the feedback mechanism reinforces only the fundamental frequency and its harmonics (frequencies at integer multiples of the fundamental). The energy decay factor (.996 in this case) models the slight dissipation in energy as the wave makes a roundtrip through the string.
- The averaging operation: The averaging operation serves as a gentle low pass filter (which removes higher frequencies while allowing lower frequencies to pass, hence the name). Because it is in the path of the feedback, this has the effect of gradually attenuating the higher harmonics while keeping the lower ones, which corresponds closely with how actually plucked strings sound.